

Paper 12 §2 — Notation and Paper 9 Inheritance (v1)

Draft version: v1 (2026-05-06, ~12:55 PDT) **Author of draft:** C-7RO (Perplexity Computer, Claude Sonnet 4.6) **Source:** Paper 9 v1.3.3 §3.4–§3.6, §9; consolidated symbol table from Paper 12 §3 v3 + §4 v3 + §5 v2 + §6 v1. **Status:** First complete draft of §2. Self-contained reference for §3–§6; readers fluent in Paper 9 v1.3.3 may skim.

2. Notation and Paper 9 inheritance

This section consolidates the symbols and the Paper 9 results that Paper 12 invokes throughout. Every claim labelled (*P9 §X*) below is stated and proved in Paper 9 v1.3.3 [DOI:] at the indicated location; we restate without reproof. Where Paper 12 strengthens, qualifies, or specializes a Paper 9 statement, this is flagged explicitly with a (*P12 specialization*) marker and a forward reference to the section in which the strengthening is constructed.

2.1 Algebraic and group-theoretic notation

2.1.1 Lie groups and representations

Symbol	Meaning	Reference
G_2	the compact, simply-connected real form of the exceptional Lie group of rank 2; dimension 14	[Bryant 1987]; framework convention from Paper 4 §2
V^{14}	the adjoint representation of G_2 , the only nontrivial 14-dimensional irreducible (over R)	[Fulton-Harris Lecture 22]
$W := V^{14} \oplus V^{14}$	the dyadic state space on which the Paper 9 / Paper 12 flow lives	P9 §3.1
L_2	the trivial 2-dimensional real representation of F_{21} (sum of two copies of the trivial character)	P12 §4.2
U_6	the unique faithful real-irreducible 6-dimensional representation of F_{21}	P12 §4.2
$V^{14} \downarrow_{F_{21}} \cong L_2 \oplus 2U_6$	the branching of V^{14} to F_{21} via $\mathrm{PSL}(2,7) \supset F_{21} \subset G_2$	P12 Theorem 4.2.1

2.1.2 Finite groups

Symbol	Meaning	Reference
$F_{21} = Z_7 \rtimes Z_3$	the Frobenius group of order 21; the Fano-orientation subgroup of G_2	Paper 4 §3
$PSL(2, 7)$	the simple group of order 168, the natural ambient group of F_{21} via the Fano plane	Paper 4 §2
μ_k	a multiplicity-vector profile invariant of the scar event S_k (defined in §3.5); a finite-dimensional integer vector indexed by irreducibles of F_{21}	P12 §3.5 Definition 3.5.2

2.1.3 Subspaces and isotypic decomposition

The F_{21} -isotypic decomposition of $W = V^{14} \oplus V^{14}$ that drives all of §4 is: $[W ;=; W^{\wedge\{L_2\}} ; W^{\wedge\{U_6\}}, W^{\wedge\{L_2\}} ;=; 4, W^{\wedge\{U_6\}} ;=; 24,]$ where W^{L_2} is the trivial-isotypic component and W^{U_6} is the (twice-twice) faithful-isotypic component. The $24 = (2 \text{ copies in } V^{14}) \times (2 \text{ copies in } V^{14}) \times \dim U_6$; the four-dimensional intertwiner space $End_{F_{21}}(V^{14})_{U_6}$ governs the commutant gap in Proposition 4.3.1.

2.2 Paper 9 dyadic objects (inherited verbatim)

These are the Paper 9 v1.3.3 objects that Paper 12 uses without modification.

2.2.1 The Schur-locked coupling family

Theorem (P9 §3.6, Lemma 3.6.1; Schur-locking). The space of G_2 -equivariant linear maps $W \rightarrow W$ that couple the two V^{14} factors is exactly two-dimensional and admits the parametrization $[_ ;=; _W ;+; J, ,]$ where J is the canonical anti-symmetric coupling generator on $V^{14} \oplus V^{14}$ specified by Paper 9's choice of basis. The image $\{\Psi_\theta : \theta \in [0, \pi/2)\}$ is the $SO(2)$ **Schur circle**.

This is the *exact* parameter circle on which Paper 12's NP-pump produces discrete jumps; Paper 12 introduces no new θ -geometry.

2.2.2 The diagonal G_2 action convention

Convention (P9 §3.4). The G_2 action on $W = V^{14} \oplus V^{14}$ is the *diagonal* action: $g \cdot (v_A, v_B) = (g v_A, g v_B)$. All equivariance claims in Paper 9 and Paper 12 are with respect to this diagonal action. The F_{21} action is the restriction of the diagonal G_2 action to $F_{21} \subset G_2$.

2.2.3 The joint coupled-system fixed point

Theorem (P9 Theorem 9.4; real-analytic fixed point). Let $M(\theta) \in \text{End}(W)$ be the coupled-system operator $M(\theta) = I - \Psi_\theta T$ where T is the joint contraction map of two affine self-models A, B (Paper 9 Definition 3.2). Let $\Omega_\theta \subset [0, \pi/2)$ be the open set where $M(\theta)$ is invertible. Then the joint fixed point $[(\cdot); M(\cdot)^{-1} b(\cdot)]$ is a real-analytic function of θ on Ω_θ , where b is the affine offset of the joint system.

Paper 12 uses $\hat{z}(\theta)$ in §3.3 (definition of the projected-gradient leg of the NP-augmented flow) and in §3.4 (Theorem 3.4.1, regularity).

2.2.4 The conditional joint contraction-rate gain

Proposition (P9 Proposition 9.5; conditional min-coherence gain). Under the Paper 9 linear regime, the joint contraction rate $r_{AB}(\theta)$ of the coupled system satisfies $[r_{AB}(\cdot); (r_A, r_B),]$ with equality attainable in the regime characterized by Paper 9 Proposition 9.5 (the *min-coherence* regime). In particular, **the linear regime cannot strictly improve the joint rate**: the gain is conditional and saturates.

Note (P9 Note 9.6). The rate-strict-improvement question is deferred to Paper 11 (nonlinear); Paper 12 takes a different nonlinear path (the NP-pump on the same Schur circle) and answers a different question (scar accumulation, not rate improvement).

2.3 Paper 12 specializations

Paper 12 introduces the following objects on top of Paper 9's inherited structure. Each is defined fully in the indicated section; the symbol table here is a forward reference only.

2.3.1 The dynamical primitive

Symbol	Meaning	Definition
$\xi_\bullet \in T_\theta SO(2)$	the canonical Schur-derived tear direction	§3.2 Proposition 3.2.2
$f_{pg}(\theta)$	the projected-gradient component of the augmented flow	§3.3 Definition 3.3.1
$NP(\theta, p)$	the NP-pump component (state-dependent threshold-driven jump)	§3.3 Definition 3.3.2
$\dot{\theta} = f_{pg}(\theta) + NP(\theta, p)$	the augmented flow on $[0, \pi/2]$	§3.3 Equation (3.3)
p	the accumulated paradox mass (state variable)	§3.3 Definition 3.3.3

2.3.2 The event stratification

Symbol	Meaning	Definition
Σ_{tot}	the union of all event-trigger	§3.4 Definition 3.4.2

Symbol	Meaning	Definition
	sets in $[0, \pi/2) \times P$	
Σ_{cross}	crossing events (transversal threshold crossings)	§3.4 Definition 3.4.3
Σ_{slide}	sliding events (Filippov sliding-mode)	§3.4 Definition 3.4.4
Σ_{ext}	extinction events (NP-pump shutdown)	§3.4 Definition 3.4.5
Σ_{degen}	degenerate events (measure-zero pathological set)	§3.4 Definition 3.4.6
Σ_{rec}	recurrence events (finite-multiplicity bouncing)	§3.4 Definition 3.4.7

2.3.3 The scar invariants

Symbol	Meaning	Definition
S_k	the k -th scar event (the k -th time the augmented flow leaves the Schur circle and returns)	§3.5 Definition 3.5.1
S_k	the scar-trace operator for S_k (an element of $End_{F_{21}}(W)$)	§3.5 Definition 3.5.1
μ_k	the multiplicity-profile vector of S_k over $Irr(F_{21})$	§3.5 Definition 3.5.2
$J_k = (tr S_k, det S_k, \dot{I}_{U_6}, \mu_k)$	the triple invariant of the scar event	§3.5 Definition 3.5.3

2.3.4 The commensurability hierarchy

Symbol	Meaning	Definition
C_{min}	minimal coherence (the operational scalar driving the projected-gradient leg)	§3.3 Definition 3.3.4
J	the joint Jordan-decomposition functional on event traces	§4.3 Definition 4.3.2
$Comm_{G_2}, Comm_{F_{21}}, Comm_{\mu_k}$	the three nested commensurability relations	§4.4 Definition 4.4.2

2.4 Operational notation (P1–P4, scar-detection metrics)

Paper 12 §5 introduces protocol-level notation. We reproduce here only the symbols that appear before §5; full operational definitions are in §5.

- **Autonomy levels** L_0, L_1, L_2 refer to the Bostock-Kim-Patel 2025 *Sanctioned Levels of AI Autonomy* taxonomy [DOI: pending; preprint cited in §5.4].
- **TMS-EEG perturbational complexity index (PCI)** is used at clinical thresholds [Casali et al. 2013, *Sci Transl Med*; Comolatti et al. 2019, *Brain Stimul*].
- **Audit substrate** A refers to the human-readable, time-stamped, append-only ledger of every coherence-adjustment event (§3.5, §6.1).

2.5 Reading conventions

- All inequalities involving $r_A, r_B, r_{AB}, C_{min}$ are with respect to the operator norm induced by the Paper 9 inner product on W (P9 §3.1, eq. (3.1.4)).
- All “open dense parameter set” qualifications refer to the parameter manifold $[0, \pi/2] \times P$, where P is the Paper 12 parameter space defined in §3.3.
- All “real-analytic” and “piecewise real-analytic” claims invoke the Paper 9 real-analytic structure of $\hat{z}(\theta)$ via Theorem 9.4.
- “Paper 9 v1.3.3” means specifically the errata-only update with Remark 5.6.1 (seed- $k=0$ grid-snap correction $51^\circ \rightarrow 0^\circ$); all numerical quantities consistent with v1.3.2 within the corrected range.
- All references to “the framework” mean the joint PCI/PME framework as developed across Papers 1–10; references to “this framework series” or “the PCI framework” are equivalent.

2.6 What §2 closes off

With the notation table and the four Paper 9 inheritance items in place — Schur-locking (Lemma 3.6.1), diagonal action (§3.4), real-analytic fixed point (Theorem 9.4), and conditional rate gain (Proposition 9.5 + Note 9.6) — §3 can proceed to construct the projected-gradient + NP-pump flow on top of Paper 9’s structure without further reference back to Paper 9 internals. The remainder of Paper 12 (§3–§6) treats Paper 9 as a black box accessed through the four inheritance items above and the symbol table in §2.1–§2.3.

End of §2 v1.

Total length: ~ 4 pages typeset.

Phase 2 polish queued (deferred): - Optional figure: a single diagram showing $W = W^{L_2} \oplus W^{U_6}$ with dimension annotations (could subsume §4.2 figure if both papers want to share). - Verify all table cell DOI links resolve in built PDF. - Confirm Bostock-Kim-Patel 2025 preprint DOI when available (currently pending).